Data Understanding and Measurement Basics

Software / Machine Learning Domain

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Repository

GitHub Repository:

https://github.com/SnigdhaRoutray/Synergy_TP

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1. Introduction

Any data set needs to be fully understood before it is analyzed or modeled. This includes what each value means, what unit it is measured in, and under what circumstances it was collected. This report provides this groundwork in the fields of Biochemistry, Electronics, and Mechanics.

2. Observation vs. Variable

Observation

An observation is a piece of information collected about a particular subject. Each observation corresponds to a record/row in a dataset.
Example: A particular student's marks in a test.

Variable

A variable is a property that can take different values for various observations. It is a column in a dataset.
Example: Score, Gender, Age etc.

3. Categorical vs. Numerical Data

Categorical Data

Categorical data is data that can be put into groups or categories using labels to classify them. There are two primary categories of categorical data:

- **Nominal data:** This is the data category that names or labels its categories. These are categories with no specific rank or order.
  Example: Gender can be classified into 3 categories: Male, Female, Other.
- **Ordinal data:** Elements with rankings, orders, or rating scales are included in this category of categorical data.
  Example: Education levels: High School, Bachelor's Degree, Master's Degree

Numerical Data

Numerical Data is the data expressed in numerical terms rather than in natural language. Numerical data can be of two types:

- Discrete Data: Countable numerical data are discrete data.
  Example: Number of students in a class
- Continuous Data: This is an uncountable data type for numbers.
  Example: Height, Weight

4. Categorical and Numerical Examples by Domain

Domain	Categorical Example	Numerical Example	Why Unit/Context Matters
Biochemistry	Sample state: fresh or degraded	Enzyme concentration, absorbance, pH	An absorbance of 1.2 means nothing without knowing the wavelength and reference blank used
Electronics	Circuit mode: idle or active	Voltage, current draw, resistance	2 A and 2 V are different physical quantities and can't be compared directly
Mechanical	Surface finish: rough or polished	Applied stress, strain, deflection	50 MPa is a stress value, not a percentage or a count

Table. 1 Domain wise examples for Categorical and Numerical data

5. Raw Data vs. Processed Data

Raw Data

Raw data is the original information collected directly from various sources before any manipulation, cleaning, or analysis takes place. Examples of raw data sources include:

- **Surveys and Questionnaires**: Responses collected from participants.
- **Transaction Logs**: Data generated from sales or service interactions.
- **Sensor Data**: Information collected from devices and sensors.
- **Social Media Interactions**: Posts, comments, and reactions on social media platforms.

Characteristics of Raw Data:

- **Unfiltered**: Raw data remains untouched and unaltered, preserving the original information as collected.
- **Varied Formats**: It can come in different forms, making it diverse and complex.
- **Large Volume**: Raw data often consists of vast amounts of information, which can be overwhelming without processing.
- **Potential Errors**: Raw data may contain inaccuracies, duplicates, or irrelevant information that need to be addressed.

Processed Data

Processed data refers to raw data that has undergone a series of transformations and analyses to make it more useful and informative. The process can include cleaning, organizing, aggregating, and interpreting the data to extract meaningful insights. Processed data is generally more structured and easier to work with, making it suitable for analysis, reporting, and decision-making. Some common processing steps include:

- **Data Cleaning:** Removing duplicates, errors, and irrelevant information.
- **Data Transformation:** Normalizing formats and standardizing values.
- **Data Aggregation:** Summarizing data to produce meaningful statistics.
- **Data Analysis:** Applying statistical or analytical methods to derive insights.

Characteristics of Processed Data:

- **Cleaned and Organized:** Processed data is refined to eliminate errors and inconsistencies, enhancing its quality.
- **Structured Format:** It often follows a specific format (like tables or databases), making it easier to analyze.
- **Informative:** Processed data is designed to convey insights, trends, and patterns that aid in decision-making.
- **Summarized Information:** It often includes aggregated data, such as averages or totals, to provide clearer insights.

6. Measurement Units and Why They Matter

A measurement unit is a standardized reference quantity used to express a measured value (volts, pascals, mol/L, newtons, etc.). A number without a unit carries no physical meaning. "5" is meaningless until you know if it's 5 volts, 5 amps, or 5 newtons. Units also allow measurements to be compared, combined, and reproduced.

7. Why Comparing Values Across Units Is Dangerous

Values that may have similar numbers can mean completely different things if they do not share the same units. For example, it is meaningless to compare 5mV and 5V because of the 1000-fold difference in scales and even more so if one is comparing pH value with voltage; they both describe two completely different aspects of nature.

8. Mean, Median, Minimum, Maximum, Range

- **Mean:** sum of all values divided by the count
- **Median:** the middle value when data is sorted in order
- **Minimum:** the smallest recorded value
- **Maximum:** the largest recorded value
- **Range:** the spread of the data, calculated as maximum minus minimum

9. When the Mean Can Be Misleading

Mean is affected by outliers and skewness. Outliers may influence the value of the mean by pulling it away from the "true" centre of the dataset. For example, one observation of 500 volts in a 4 to 6 volt dataset will shift the mean significantly.

10. What to Inspect Before Analyzing a Dataset

Before performing any analysis, it is important to look out for Missing or null values, Inconsistent or missing units, Data type issues, Duplicate entries, Outliers and other obviously wrong readings, Sample size, and the general shape of the distribution.

11. Questions to Ask Before Analyzing Data From Another Team

- What type of instrument/measurement technique was used to obtain the data and was it calibrated?
- What are the units of measure for each column?
- What were the conditions (temp, load, etc.) under which the data was obtained?
- Is there a data dictionary or column descriptions available?
- Has there been any pre-processing applied (smoothing, averaging, etc.)?
- What is the known error margin on the measurements?
- Are there any known gaps, missing data or outliers?

12. Domain Thinking: A Dataset Is Not Just a Table

For the numbers in the spreadsheet to hold any meaning, it needs to be connected with four components which include; variable name, measurement units, experiment conditions, and context. The number “120” means nothing by itself. “120N of tensile force on Material A at 25 degrees C” now makes sense as data. This is why it is important to understand data structures before we carry out statistical analyses or apply machine learning.

References

[1] Khan Academy, "Statistics and Probability," Khan Academy. [Online]. Available: https://www.khanacademy.org/math/statistics-probability

[2] D. Diez, M. Çetinkaya-Rundel, and C. Barr, *OpenIntro Statistics*. [Online]. Available: https://www.openintro.org/book/os/

[3] National Institute of Standards and Technology, "Exploratory Data Analysis," *NIST/SEMATECH e-Handbook of Statistical Methods*. [Online]. Available: https://www.itl.nist.gov/div898/handbook/eda/eda.htm

[4] QuestionPro, "Categorical Data vs. Numerical Data: Differences and Examples," *QuestionPro Blog*. [Online]. Available: https://www.questionpro.com/blog/categorical-data-vs-numerical-data/.